



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
15-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

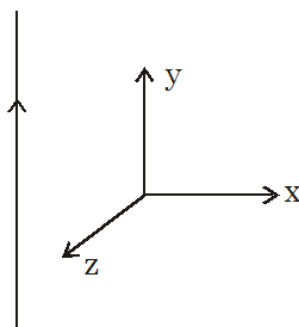
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

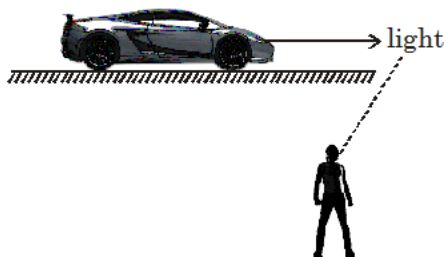
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Consider an infinite wire carrying current $i = \frac{i_0 t}{T}$, Here i_0 & T are constant, t is time. The wire is in xy plane as shown. At the origin,

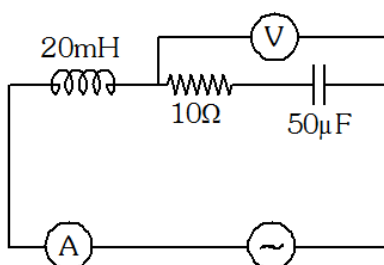


- (A) Displacement current is in +ve y -direction.
- (B) Electric field is in +ve z -direction.
- (C) Poynting vector $\frac{(\vec{E} \times \vec{B})}{\mu_0}$ is in +ve x -direction.
- (D) Magnetic field is in -ve z -direction.

2. Light coming from an automobile headlights is normally polarized such that plane of electric field is vertical. This light strikes dust particles in air & is scattered. Suppose an observer is standing $\perp$ to road as shown. The light that he receives is only scattered light. The light is :



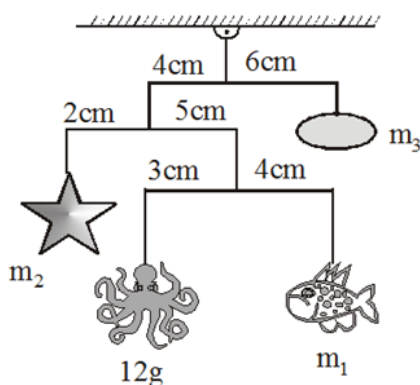
- (A) Having no intensity in direction of observer.
 (B) Has electric field in vertical plane $\perp$ to car's velocity.
 (C) Has magnetic field parallel to car's velocity but in horizontal plane.
 (D) Has magnetic field in vertical plane parallel to car's velocity.
3. A coil of radius 1 cm carries a current of 1A. The magnetic field due to the coil :-
 (A) on its axis at a distance of 1m from centre is approximately 0.2 nT.
 (B) in the plane of coil at a distance of 1m from centre is approx 0.1 nT
 (C) is same at 2 points on the axis equidistant from the centre.
 (D) is maximum at the centre compared to any other point on the axis.
4. For the AC circuit shown, the reading of ammeter and voltmeter are 5A and $50\sqrt{5}$ volts respectively, then :



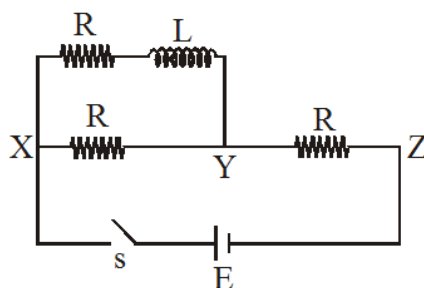
- (A) average power delivered by the source is 250W
 (B) rms value of AC source is 50 volts
 (C) rms value of AC source is 100 volts
 (D) frequency of AC source is $1000/2\pi$ Hz

Space for Rough Work

5. A fancy showpiece is constructed of light rods, light strings, and small toys, as shown below. Determine the masses of the objects m_1 , m_2 , m_3 .



- (A) $m_1 = 9g$ (B) $m_2 = 52.5g$ (C) $m_3 = 105g$ (D) $m_2 = 49g$
6. The switch is closed at $t = 0$ in the adjoining circuit. Which of the following is/are correct ?



- (A) The potential difference across YZ at $t = 0$ is $2E/3$.
- (B) The potential difference across XY at $t = \infty$ is $E/2$.
- (C) The potential difference across YZ at $t = 0$ is $E/2$.
- (D) The potential difference across XY at $t = \infty$ is $2E/3$.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

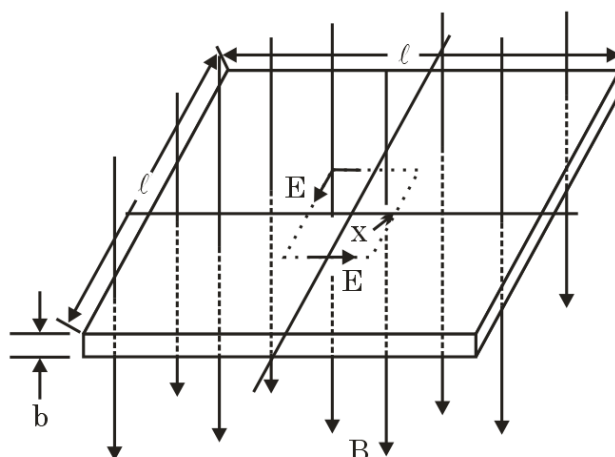
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

The figure shows a slab of metal of dimensions $\ell \times \ell \times b$, through which pass lines of uniform magnetic field directed perpendicular to the square faces of the slab. Although uniform in space, the field varies in time; its magnitude is $B = B_0 \sin \omega t$.



Consider a square wire loop of side x and infinitesimal thickness as shown in the figure. One can find the power dissipated in the loop. This is the power dissipated due to eddy currents. If the conductivity of the material is σ , find the total power dissipated in the square block by integrating the power dissipated over different square loops.

7. The average power dissipated in the loop is:

(A) $P = \frac{1}{64} \sigma \omega^2 B_0^2 b \ell^4$

(B) $P = \frac{1}{32} \sigma \omega^2 B_0^2 b \ell^4$

(C) $P = \frac{1}{48} \sigma \omega^2 B_0^2 b^2 \ell^3$

(D) $P = \frac{1}{24} \sigma \omega^2 B_0^2 b \ell^3$

8. If we divide the entire square slab into 16 equal square slabs, which are insulated from one another, the power dissipated would

(A) decrease by factor of 16

(B) decrease by a factor of 32

(C) decrease by factor of 4

(D) decrease by factor of 64

Space for Rough Work

Paragraph for Questions 9 and 10

Consider a plane electromagnetic wave whose magnetic field is given by :

$$\vec{B} = 10^{-8} \text{T} \hat{j} \sin(3\pi x + 4\pi z + \omega t)$$

The wave is travelling in free space.

9. Angular frequency ω is :

- (A) $9\pi \times 10^8 \text{ rad/s}$ (B) $12\pi \times 10^8 \text{ rad/s}$ (C) $15\pi \times 10^8 \text{ rad/s}$ (D) $\pi \times 10^8 \text{ rad/s}$

10. The electric field is given by :

- (A) $\vec{E} = (2.4\hat{i} + 1.8\hat{k}) \sin(3\pi x + 4\pi z + \omega t)$ (B) $\vec{E} = (-2.4\hat{i} + 1.8\hat{k}) \sin(3\pi x + 4\pi z + \omega t)$
 (C) $\vec{E} = (2.4\hat{i} - 1.8\hat{k}) \sin(3\pi x + 4\pi z + \omega t)$ (D) $\vec{E} = (-2.4\hat{i} - 1.8\hat{k}) \sin(3\pi x + 4\pi z + \omega t)$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

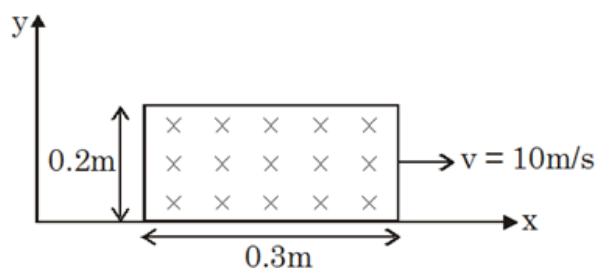
Paragraph for Questions 1 and 2

A body of mass 10 gm performs damped oscillation with a maximum amplitude of 7 cm, zero initial phase and damping coefficient of 1.6 / sec. It is acted on by an external periodic force which produces forced oscillation $x = (5 \text{ cm}) \sin (10\pi t + \frac{3\pi}{4})$.

1. The maximum magnitude of periodic force (in 10^{-2}N) is :
2. Q value of damped oscillation is :

Paragraph for Questions 3 and 4

In space, magnetic field is everywhere $\perp$ to plane of paper & directed away from the reader. The magnetic field is $B = \alpha x$, $\alpha = 0.2 \text{ T/m}$. A rectangular loop of side 0.2 m and 0.3 m is in xy plane and is being pulled with a constant speed of 10 m/s as shown. Resistance of loop is 0.8Ω .



3. The magnitude of induced current in the loop is ____ A.
4. The rate at which work is done by the external agent is _____ mW.

Space for Rough Work

Paragraph for Questions 5 and 6

Kinetic inductance : When we calculate inductance of an inductor, it is representing the tendency of current through the inductor coil to remain constant. This is due to faraday's law. In addition, electrons in the coil have inertia, so they also tend to move with a constant speed. So total inductance of an inductor is sum of both these effects. This kinetic inductance can be found by energy considerations. Consider an ideal solenoid of cross section radius 1 cm, length of 10m, area of cross section of it's wires = 1 mm^2 , no. of turns/ length = 1000, $n = 10^{28} / \text{m}^3$.

5. Ratio of kinetic inductance and inductance is of order of 10^n . Find n.
6. If current $i = i_0 \sin \omega t$, what is ω (in rad/s) at which the peak emf due to kinetic inductance becomes 1V ?
Take $i_0 = 1000 \text{ A}$. Find $\left(\frac{\omega}{10^8} \right)$.

Space for Rough Work

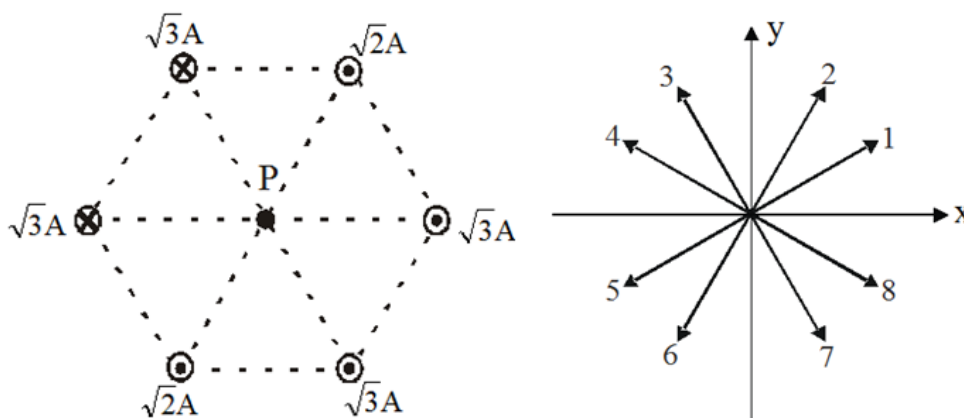
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

- A linear harmonic oscillator is subjected to a periodic force such that it's amplitude is maximum at an angular frequency of 10 rad/s. What is the phase difference between applied force and it's velocity at this frequency ? It's damping coefficient is 0.1 / sec. Fill the value of $|\cot \phi|$.
- Current $i = 2.5$ A flows along the circular coil whose equation is given by $x^2 + y^2 = 9$ cm² (here x & y are in cm). If magnetic field is given by $\pi a \times 10^{-9}$ T at point $(0, 0, 4$ cm). State value of a .
- Consider a set of six infinite long straight parallel wires arranged perpendicular to the plane of paper in a hexagon as shown. The length of the each side of the hexagon is 3cm. What is the magnitude and direction of the magnetic field at point P? To express your answer, use the following method. Two directions symmetrically in each quadrant have been shown. If the strength of magnetic field is $70\mu\text{T}$, directed along vector numbered 8, write your answer as 870. Similarly if the strength of magnetic field is $70\mu\text{T}$, directed along vector numbered 3, write your answer as 370.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

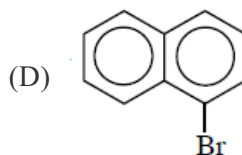
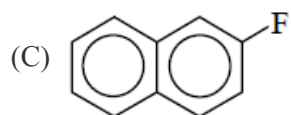
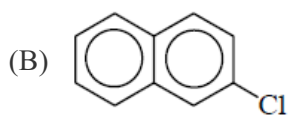
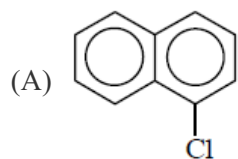
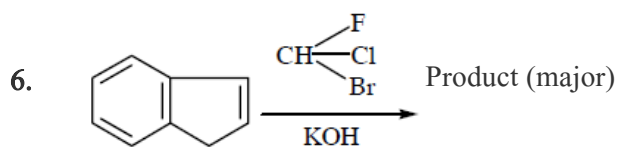
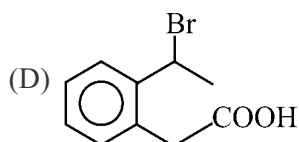
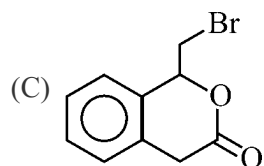
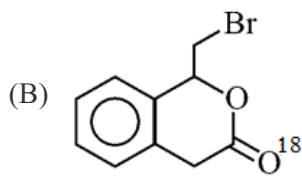
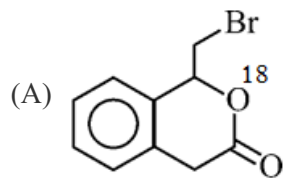
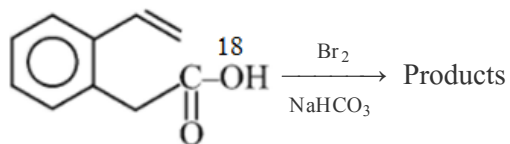
1. For equilibrium, $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$, on increase in temperature -
- (A) K_{eq} decreases because increase in k_b is more than increase in k_f
- (B) K_{eq} decreases because increase in k_b is more than decrease in k_f
- (C) K_{eq} decreases because favourable $\Delta S_{\text{surr.}}$ decreases.
- (D) K_{eq} decreases because favourable $\Delta S_{\text{surr.}}$ increases.

Space for Rough Work

2. During expansion of ideal gas -
- (A) Entropy of system may increase, decrease or remain constant if process is reversible.
(B) Entropy of system may increase decrease or remains constant if process is irreversible.
(C) Entropy of isolated system may increase decrease or remains constant if process is irreversible
(D) Entropy of system at constant U & V always increases if process is irreversible.
3. Which of the following complex(es) may show secondary valency by Chloride/Bromide ion?
- (A) $\text{NiCl}_2 \cdot 6\text{NH}_3$ (B) $\text{CoBrSO}_4 \cdot 5\text{NH}_3$
(C) $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ (D) $\text{PdCl}_2 \cdot 2\text{HCl}$
4. Which of the following is/are bidentate ligand?
- (A) en (B) dien (C) acac^{-1} (D) gly^{-1}
-

Space for Rough Work

5. The product(s) of the given reaction is/are :



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

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- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
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Paragraph for Questions 7 and 8

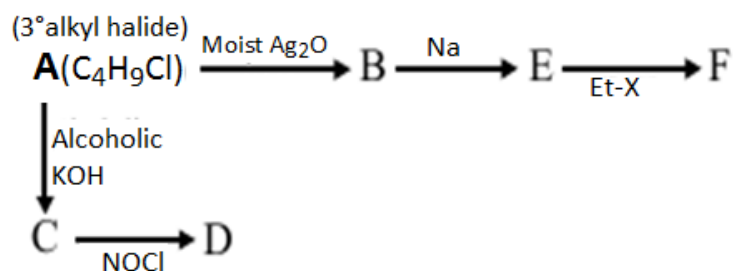
Radioactive tracers provide a convenient mean of measuring the small concentration encountered in determining K_{sp} values. Exactly 30 ml of 0.001M AgNO_3 solution containing radioactive silver, with an activity of 40000 counts per ml was mixed with 10 ml of 0.002 M KIO_3 solution and the mixture is diluted to exactly 400 ml. After equilibrium was reached, a portion of sample was filtered to remove any solid and was found to have an activity of 2000 counts per ml.

7. The K_{sp} of AgIO_3 is -
- (A) 6×10^{-10} (B) 1.25×10^{-9} (C) 6×10^{-12} (D) 4×10^{-10}
8. Number of moles of solid AgIO_3 formed -
- (A) 10^{-5} (B) 10^{-4} (C) 10^{-2} (D) 0.4×10^{-3}

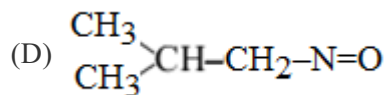
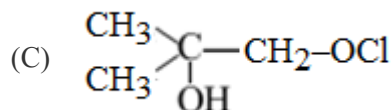
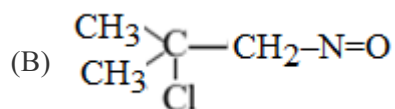
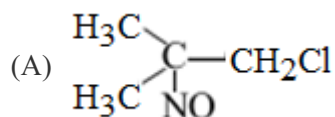
Space for Rough Work

Paragraph for Questions 9 and 10

Consider the following reaction sequence :



9. The correct structure of D is -



10. Identify 'F' from the following -



Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

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- There are **TWO (02)** questions corresponding to each question stem.
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- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
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Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

100ml of 1M Na_3PO_4 is titrated with 1M HCl the following observation were made

Stage of titration	pH
I half equivalence point	12
I equivalence point	10
II equivalence point	6

1. pH at III equivalence point should be ($\log 5 = 0.7$)
2. pH at II half equivalence point will be -

Space for Rough Work

Paragraph for Questions 3 and 4

The metal–carbon bond in metal carbonyls posses both σ & π character. The metal to ligand bonding creates a synergic effect which strengthens the bond between CO and the metal. Due to synergic effect M–C bond length & C–O bond length affected.

3. Consider $[\text{Co}(\text{CO})_4]^-$ as x and $[\text{Mn}(\text{CO})_6]^+$ as y then find how many statements are correct?

- (i) The C – O bond length in x is more than that in y
- (ii) The C – O bond energy in x is more than that in y
- (iii) Double bond character in metal to carbon bond in x has more than that in y
- (iv) Both x and y have the same EAN of the central metal atom

4. Consider the following species

$[\text{Ni}(\text{CO})_4]$, $[\text{V}(\text{CO})_6]^-$, CO^+ , CO_2 ,
 H_2CO_3 , CH_3COO^- , $[\text{NiP}(\text{CH}_3)_3(\text{CO})_3]$,
 $[\text{Cr}(\text{CO})_6]$, $[\text{NiPF}_3(\text{CO})_3]$

If bond length in CO is 'd' in isolated molecule.

The number of species having higher C-O bond length then 'd' is/are 'X'

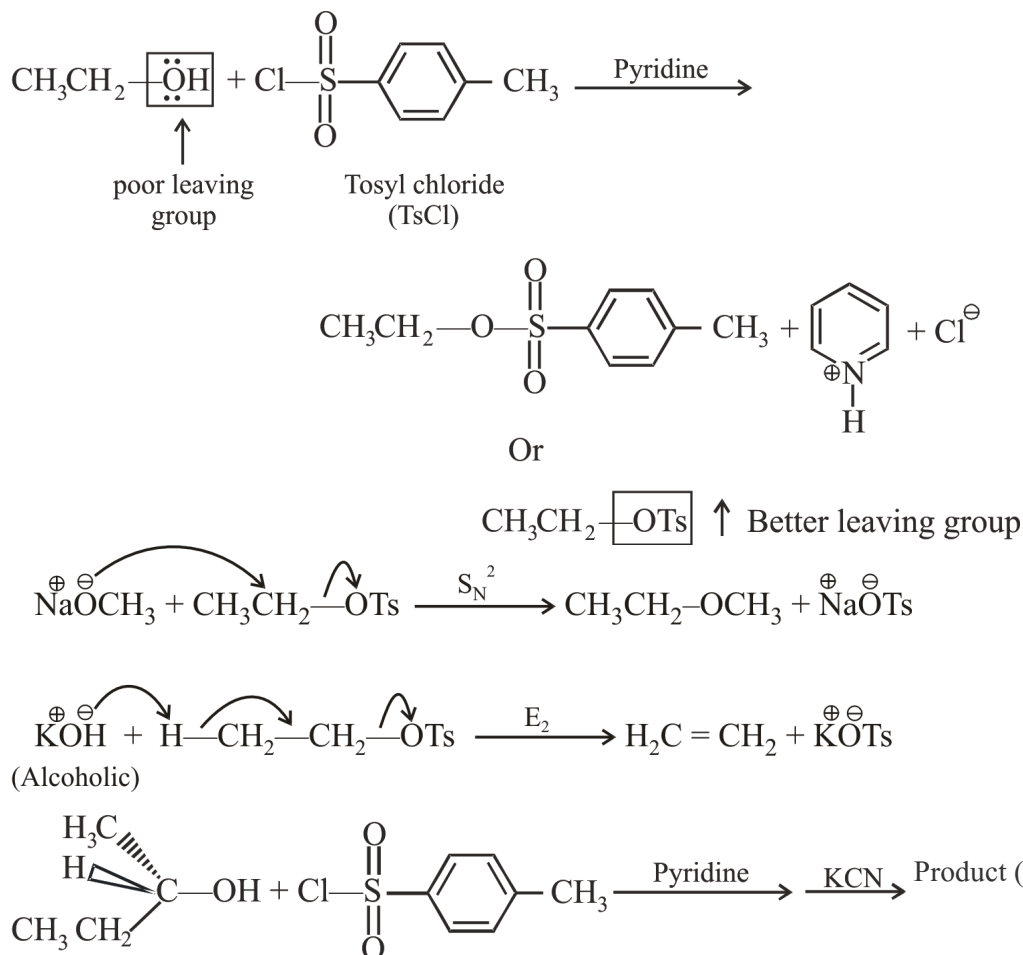
The number of species having lower C-O bond length then 'd' is/are 'y'

Find the value of $\left(\frac{X}{Y}\right)$

Space for Rough Work

Paragraph for Questions 5 and 6

Alcohols are converted to tosylates by treatment with p-toluene sulfonyl chloride (TsCl) in presence of pyridine. This overall process converts a poor leaving group ($\text{OH}^\ominus$) into better one ($\text{OTs}^\ominus$). A tosylate is a better leaving group because its conjugated acid p-toluene sulfonic acid is strong acid. Because alkyl tosylates have better leaving tendency they undergo both nucleophilic substitution and β -elimination.



- In IUPAC naming, locant of side-chain in product (X)
- Stereoisomers of product (X) is?

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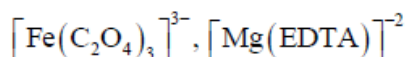
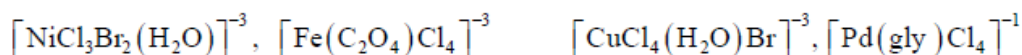
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. α -particle (${}^4_2\text{He}^{++}$) is moving along x-axis with 0.4 meV. If uncertainty in velocity is 1 % and minimum uncertainty in x-co-ordinate is 1.80×10^{-x} m, then x is -
8. For the organometallic complex compound "Ferrocene" consider,
 x – Atomic Number of its central metal atom
 y – Oxidation number of its central metal atom
 z – Number of electrons provided by the ligand for sharing with its central metal atom
 Calculate $x - y + z$
9. Consider following complexes for stereo isomerism :



Let number of complexes showing

- (i) Only geometrical isomerism = x
 (ii) Only optical isomerism = y
 (iii) neither geometrical nor optical isomerism = z
 (iv) Showing both geometrical and optical isomerism = t

Find $\left(\frac{xy}{z} - t \right)$

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

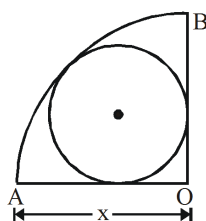
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $\frac{\cos^4 x}{\theta_1} + \frac{\sin^4 x}{\theta_2} = \frac{1}{(\theta_1 + \theta_2)}$, then $\frac{d\theta_2}{d\theta_1}$ equals (where x is not dependent on θ_1 or θ_2)

- (A) $\sin^2 x$ (B) $\cos^2 x$ (C) $\tan^2 x$ (D) $\sec^2 x$

2. Sector OAB is a quarter of circle of radius x. A circle of radius 'r' is drawn inside this sector tangent to three points as shown, then which of the following is/are true ? (where r & x are variable)



(A) $\left(\frac{dr}{dx}\right)^2 + \left(\frac{dx}{dr}\right)^2 + 2\frac{d^2x}{dr^2} = 6$

(B) $\left(\frac{dr}{dx}\right)^2 + \left(\frac{dx}{dr}\right)^2 - 2\frac{d^2x}{dr^2} = 8$

(C) $\frac{x+r}{x-r} + \frac{r-x}{x+r} = 2$

(D) $\sum_{n=1}^{10} \frac{d^n r}{dx^n} = \frac{r}{x}$

Space for Rough Work

3. Let $f(x) = \begin{cases} -2 & \text{for } -3 \leq x \leq 0 \\ x-2 & \text{for } 0 < x \leq 3 \end{cases}$, $g(x) = [f(|x|)] + [f(x)]$ where $[.]$ denotes the greatest integer function.

Which of the following statement(s) is/are true ?

- (A) $g(1.7) = -1$ (B) $g(x)$ is not differentiable at any point
(C) $g(x)$ is continuous at all non integral points (D) $g(x)$ is differentiable at some points
4. A right angled triangle has one leg of length 1, another of length 'x' & hypotenuse of length 'y'. The angle opposite to the side x is θ then -

- (A) $y^n - x^n = \frac{1 - \sin^n \theta}{\cos^n \theta}$ (B) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y - x) = 0$
(C) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^2 - x^2) = 1$ (D) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^3 - x^3)$ does not exist.

5. Given f and g are twice differentiable functions and $f' \neq 0$, $g' \neq 0$. If $f(x)g(x) = 1$, then $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)}$ is equal to :

- (A) $\frac{-2g'(x)}{g(x)}$ (B) $\frac{g'(x)}{g(x)}$ (C) $\frac{2f'(x)}{f(x)}$ (D) $\frac{-f'(x)}{f(x)}$

6. The hands of an accurate clock have lengths 3 and 4. The distance L between the tips of the hands is increasing most rapidly, when L is equal to

- (A) 5 (B) $\sqrt{7}$ (C) 25 (D) $\sqrt{2}$

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SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Consider $f(x) = \left[\ln \left(\sin x + \sqrt{\sin^2 x + 1} \right) \right] + \left[\ln \left(\sqrt{\sin^2 x + 1} - \sin x \right) \right]$

$g(x) = \left\{ \frac{x}{\pi} \right\} - \left[\frac{\sin^2 2x + \cos^4 2x}{\cos^2 2x + \sin^4 2x} \right]$ and $h(x) = f(x) + g(x)$

(where $[.]$ and $\{.\}$ are greatest integer and fractional part function respectively), then

7. Which of following is correct -

(A) Equation $f(x) = g(x)$ has infinite many solutions

(B) Period of $f(x)$ is not defined

(C) $\lim_{x \rightarrow 0} f(x)$ does not exist

(D) $\lim_{x \rightarrow 0} f(x) \cdot g(x)$

8. Number of points in $[-2\pi, 2\pi]$ where $h(x)$ is not differentiable, is -

(A) 2

(B) 3

(C) 4

(D) 5

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Paragraph for Questions 9 and 10

Let $f(x)$ and $g(x)$ be two quadratic polynomials $f(x) = ax^2 + bx + c$, $g(x) = dx^2 + ex + f$ where $a, b, c, d, e, f \in \mathbb{R}$

9. If a, b, c are the positive real numbers and $f(x) \geq 0 \forall x \in \mathbb{R}$, then the minimum value of $\left(\frac{a+b+c}{b}\right)$ is
 (A) 1 (B) 2 (C) 3 (D) 0
10. If $H(x) = f(x) \cdot g(x)$ where $ad = 1$ and $H(1) = 1, H(2) = 2, H(3) = 3$, then $\left\{ \frac{H(-1) + H(5)}{H(0) + H(4)} \right\} =$
 (where $\{ \}$ represents fractional function) is
 (A) $\frac{5}{7}$ (B) $\frac{2}{7}$ (C) $\frac{3}{7}$ (D) $\frac{4}{7}$

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SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let $f(x) = \tan\left(\frac{\pi(x+1)}{8}\right) + a$ & $g(x) = x^2 + bx + 1$. If $h(x) = \lim_{n \rightarrow \infty} \frac{f(x) + (x^2)^n g(x)}{1 + (x^2)^n}$ is continuous at $x = \pm 1$ then

1. $a^2 + b^2$ is equal to
2. Minimum value of $g(x)$ is

Paragraph for Questions 3 and 4

Let $f(x) = \cos^{-1}\left(\ln\left(\frac{x+1}{x}\right)\right)$, then

3. The number of integers that does NOT lie in the Domain of $f(x)$, is
4. The number of integers lying in the range of $f(x)$, is

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Paragraph for Questions 5 and 6

Let $g(x) = 6x^2 - 18x + 8$;

$f_1(x) = |g(x)|$, $f_2(x) = |f_1(x) - P_1|$, $f_3(x) = |f_2(x) - P_2|$, $f_4(x) = |f_3(x) - P_3|$.

5. If $P_1 = 7$, then the largest integral value of P_2 such that $f_3(x)$ has exactly 10 points of non-differentiability, is
6. If P_1, P_2, P_3 are consecutive primes such that $5 < P_1 < P_2 < P_3$ and total number of non-differentiable points of f_1, f_2, f_3 are a_1, a_2, a_3 respectively, then the number of solution of equation $H(x) = \frac{(x - a_1)(x - a_3)}{(x - a_2)} = 1$, is

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SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Let $f(x) = ax^9 + b \sin x + cx^2 \operatorname{sgn}(x) + \frac{e^x - e^{-x}}{e^x + e^{-x}}$ be defined on set of real numbers (where $a > 0, b, c \in \mathbb{R}$).
If $f(-5) = 5, f(-2) = -3$, then the minimum number of zeroes of the equation $f(x) = 0$, is :
(Where $\operatorname{sgn}(x)$ denotes signum function of x)
8. If $f(x) = \begin{cases} \frac{\sin^{-1}(\{x\}^2 + \{x\}^3) - x}{\{x\}}, & x \neq 0 \\ -1, & x = 0 \end{cases}$, then the absolute value of RHD of $f(x)$ at $x = 0$, is (where $\{.\}$ denotes fractional part function)
9. If θ is the acute angle between the curves $y = |x^2 - 1|$ and $y = |x^2 - 3|$ at their point of intersection when $x > 0$, then $49 \tan^2 \theta$ is equal to

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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